Image Fragment Reconstruction via Adjacency Prediction

Case Study Report

1. Model Architecture

The model is a **Siamese CNN** that operates on pairs of 16×16 fragments. A shared encoder maps each fragment to a 256-dimensional embedding; a single comparison head maps the two embeddings to a scalar confidence score $p_{ij} \in [0, 1]$.

Encoder. Three Conv–BatchNorm–ReLU blocks with max-pooling after the last two reduce the spatial dimensions from 16×16 to 4×4; a linear layer projects the flattened features to 256-d (ReLU + dropout $p = 0.3$).

Comparison head. Inspired by InferSent [1], the head takes the 1024-dimensional vector $[\mathbf{a} - \mathbf{b} \parallel \mathbf{a} \odot \mathbf{b} \parallel \mathbf{a} \parallel \mathbf{b}]$ as input and produces a scalar confidence score via one hidden layer (ReLU).

Why Siamese? Weight sharing ensures the comparison is symmetric and forces the encoder to learn a universal fragment representation rather than one tied to a specific input slot. No image-level labels are needed: the only supervision is whether two fragments are adjacent, which follows directly from their grid positions.

Why adjacency prediction? We preferred this over contrastive approaches because the adjacency signal is local and geometrically interpretable — each label has a direct spatial meaning — which makes failure modes transparent and mirrors the edge-continuity cues humans use when solving jigsaw puzzles. This made it easier to reason about bottlenecks and diagnose where the model struggled.

2. Training Strategy

Pretext task. The model is trained to predict spatial adjacency: $y_{ij} = 1$ iff fragments i and j are grid-neighbours within the same image. With $N = 10$ images on a 4×4 grid, the negative-to-positive ratio is exactly 12,480/240 = 52. We use weighted BCE with tilt parameter β:

$$\mathcal{L}_{\mathrm{WBCE}}(\beta) = -\tfrac{1}{N}\sum_{(i,j)} \big[\beta \cdot 52\, y_{ij} \log p_{ij} + (1 - y_{ij}) \log(1 - p_{ij})\big].$$

Multi-task loss. Adjacency is a local signal; the downstream task is global clustering. A same-image loss term $\mathcal{L}_{\mathrm{BCE}}^{\mathrm{same}}$ is added to bridge this gap, acting on the same score p_{ij}:

$$\mathcal{L} = \lambda_{\mathrm{adj}}\, \mathcal{L}_{\mathrm{WBCE}}(\beta) + \lambda_{\mathrm{same}}\, \mathcal{L}_{\mathrm{BCE}}^{\mathrm{same}}.$$

Inference. To assign each fragment to its source image, the pairwise scores must be turned into hard cluster assignments. The 160×160 matrix of scores p_{ij} is used directly as a similarity matrix for **balanced spectral clustering**: spectral embedding finds clusters as well-connected subgraphs of the similarity graph, while the Hungarian-assignment balanced k-means step enforces exactly 16 fragments per cluster, respecting the fact that every source image contributes the same number of patches. Each inference call processes one batch of 10 images ($N = 160$ fragments): N encoder passes followed by $\binom{N}{2} = 12{,}720$ head evaluations ($\mathcal{O}(N^2)$) and an $\mathcal{O}(N^3)$ spectral decomposition. At this fixed scale both steps complete in under a second.

Training details. 1.28M training images (ImageNet-64); each step samples 10 fresh images. Up to 25,000 iterations; early stopping with patience 10, evaluated every 250 steps over 20 validation batches. All runs on CPU (Apple M-series).

3. Evaluation Results

Performance is measured by the **Adjusted Rand Index** (ARI), which counts pair-level agreement between predicted and true clusters, adjusted for chance. ARI = 0 is random; ARI = 1 is perfect.

Validation results. A five-seed study (seeds 0–4) across five hyperparameter configurations (Table 1) shows that configurations (a)–(d) cluster tightly in the range 0.53–0.56 ($\sigma \approx 0.01$–0.02), while configuration (e) ($\beta = 1$, class-balanced weighting) sits noticeably lower at 0.46. A one-way ANOVA (not assuming equal variances) finds a statistically significant difference ($\alpha = 0.10$), driven by configuration (e); the loss weights within configurations (a)–(d) are not a strong lever at this scale.

Test results. The best checkpoint (run `multitask_10_seed_2`) was evaluated on 1,000 independent test batches: $\mathrm{ARI} = 0.521 \pm 0.104$, $\mathrm{NMI} = 0.686 \pm 0.071$, purity $= 0.694 \pm 0.076$, adjacency F1 = 0.356, AUROC = 0.948. The ± values are standard deviations across the 1,000 test batches for this single checkpoint, reflecting batch-to-batch variability in which images are drawn — not variance across training seeds

Configuration	Mean ARI	SD
(a) adj-only, $\beta = 1/52$	0.538	0.016
(b) same-only	0.541	0.018
(c) multi-task $\lambda_s = 1.0$	0.557	0.009
(d) multi-task $\lambda_s = 0.2$	0.531	0.012
(e) adj-only, $\beta = 1$	0.460	0.019

Table 1: Validation ARI (5 seeds each).

(reported in Table 1). In concrete terms, a purity of 0.694 means that on average approximately 5 out of 16 fragments per cluster are assigned to the wrong source image.

Pretext task. Adjacency AUROC sits at 0.95–0.97 across all configurations, indicating the model has nearly solved the binary pretext task. The remaining gap to perfect clustering is consistent with the fragment-ambiguity cases discussed in Section 4.

4. Challenges and Improvements

Class imbalance. The 52:1 negative-to-positive ratio in adjacency pairs requires careful loss weighting. The multi-seed study shows that the textbook class-balanced weight ($\beta = 1$) degrades ARI by $\sim$0.08 relative to plain BCE ($\beta = 1/52$), suggesting that aggressive upweighting forces the model to chase ambiguous positive pairs at the cost of overall clustering quality.

Local vs. global signal. Adjacency is a nearest-neighbour signal; two fragments from opposite corners of the same image are labelled negative. The same-image loss term partially closes this gap but does not produce a statistically significant improvement at this scale, suggesting the binding constraint is fragment ambiguity rather than the loss formulation.

Fragment ambiguity. The failure-modes analysis shows that near-uniform or repetitive-texture patches produce similar embeddings regardless of source image, and are frequently mis-assigned.

5. Next Steps

Pretrained encoder. Replacing the trained-from-scratch CNN with a self-supervised encoder (e.g. DINOv2) would provide richer features at 16×16 resolution at no labelling cost, likely the highest-impact single change.

Topology-aware clustering. The true clustering is more constrained than "equal sizes": each cluster's induced subgraph must be isomorphic to a 4×4 grid. A GNN-based approach that exploits this planarity constraint could resolve borderline assignments that spectral clustering misses.

Larger-scale statistical study. With $n = 5$ seeds the ANOVA is underpowered for effect sizes below $\sim$0.07 ARI. A 20-seed study would give $\Delta_{\min} \approx 0.03$, sufficient to detect the differences suggested by the single-seed sweep.

Color-histogram baseline. A non-learned baseline using per-fragment color histograms with spectral clustering would anchor the model's ARI against a trivial reference and quantify how much benefit comes from learned representation versus color matching.

References

[1] Conneau et al. (2017). *Supervised Learning of Universal Sentence Representations from Natural Language Inference Data.* EMNLP.

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